

AAI_500_M1_Assignment

September 7, 2026

1 Assignment 1.1

```
[ ]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

This is formatted as code

Name: Justin Date: Nguyen

For this assignment, you will refer to the textbook to solve the practice exercises. **Use Python to answer any coding problems (not R, even if indicated in your textbook).** Use Jupyter Notebook, Google Colab, or a similar software program to complete your assignment. Submit your answers as a **PDF or HTML** file. As a best practice, always label your axes and provide titles for any graphs generated on this assignment. Round all quantitative answers to 2 decimal places.

1.1 Problem # 1.1.

In the 2018 election for Senate in California, a CNN exit poll of 1882 voters stated that 52.5% voted for the Democratic candidate, Diane Feinstein. Of all 11.1 million voters, 54.2% voted for Feinstein.

(a) What was the (i) subject, (ii) sample, (iii) population?

(a) Your answer goes here

- i. voter
- ii. voters in California that completed the survey
- iii. Population of California

1.2 Problem # 1.2.

The **Students** data file at <http://stat4ds.rwth-aachen.de/data/Students.dat> responses of a class of 60 social science graduate students at the University of Florida to a questionnaire that asked about *gender* (1 = female, 0 = male), *age*, *hsgpa* = high school GPA (on a four-point scale), *cogpa* = college GPA, *dhome* = distance (in miles) of the campus from your home town, *dres* = distance (in miles) of the classroom from your current residence, *tv* = average number of hours per week that you watch TV, *sport* = average number of hours per week that you participate in sports or have other physical exercise, *news* = number of times a week you read a newspaper, *aids* = number of people you know who have died from AIDS or who are HIV+, *veg* = whether you

are a vegetarian (1 = yes, 0 = no), *affil* = political affiliation (1 = Democrat, 2 = Republican, 3 = independent), *ideol* = political ideology (1 = very liberal, 2 = liberal, 3 = slightly liberal, 4 = moderate, 5 = slightly conservative, 6 = conservative, 7 = very conservative), *relig* = how often you attend religious services (0 = never, 1 = occasionally, 2 = most weeks, 3 = every week), *abor* = opinion about whether abortion should be legal in the first three months of pregnancy (1 = yes, 0 = no), *affirm* = support affirmative action (1 = yes, 0 = no), and *life* = belief in life after death (1 = yes, 2 = no, 3 = undecided). You will use this data file for some exercises in this book.

- (a) Practice accessing a data file for statistical analysis with your software by going to the book's website and copying and then displaying this data file.

(a) Your answer goes here

```
[ ]: df = pd.read_csv('https://stat4ds.rwth-aachen.de/data/Students.dat', sep=r'\s+')
df.head()
```

```
[ ]:  subject  gender  age  hsgpa  cogpa  dhome  dres    tv  sport  news  aids  \
0         1         0   32    2.2    3.5     0    5.0   3.0     5    0    0
1         2         1   23    2.1    3.5   1200    0.3  15.0     7    5    6
2         3         1   27    3.3    3.0   1300    1.5   0.0     4    3    0
3         4         1   35    3.5    3.2   1500    8.0   5.0     5    6    3
4         5         0   23    3.1    3.5   1600   10.0   6.0     6    3    0

      veg  affil  ideol  relig  abor  affirm  life
0     0     2     6     2     0     0     1
1     1     1     2     1     1     1     3
2     1     1     2     2     1     1     3
3     0     3     4     1     1     1     2
4     0     3     1     0     1     0     2
```

- (b) Using responses on *abor*, state a question that could be addressed with (i) descriptive statistics, (ii) inferential statistics.

Your answer goes here

```
[ ]: abor = df['abor']
print(abor)
```

```
0    0
1    1
2    1
3    1
4    1
5    1
6    1
7    1
8    1
9    0
10   0
11   1
```

12	1
13	1
14	1
15	1
16	0
17	1
18	1
19	1
20	1
21	1
22	0
23	1
24	1
25	1
26	1
27	1
28	1
29	1
30	1
31	0
32	1
33	1
34	1
35	0
36	1
37	1
38	1
39	1
40	0
41	1
42	0
43	1
44	1
45	1
46	1
47	0
48	1
49	1
50	0
51	0
52	1
53	1
54	1
55	0
56	1
57	1
58	1
59	1

Name: `abor`, dtype: `int64`

- i. What is the percentage of the sample who think that abortion should be legal in the first 3 months of pregnancy? What's the percentage who think otherwise?
- ii. Based on the data, what percentage of the wider student population agree that abortion should be legal within the first 3 months of pregnancy and what percentage disagree?

1.3 Problem # 1.3.

Identify each of the following variables as categorical or quantitative: (a) Number of smartphones that you own; (b) County of residence; (c) Choice of diet (vegetarian, nonvegetarian); (d) Distance, in kilometers, commute to work

Your answer goes here

- a. quantitative
- b. categorical
- c. categorical
- d. quantitative or can be categorical depending on how the question is set up i.e. is it enter in the distance answer box or is it a dropdown with options? $< 10\text{mi}$, $>25\text{ mi}$, etc.

1.4 Problem # 1.4.

Give an example of a variable that is (a) categorical; (b) quantitative; (c) discrete; (d) continuous

Your answer goes here

- a. Favorite type of music
- b. Number of siblings
- c. Number of siblings
- d. Miles on car

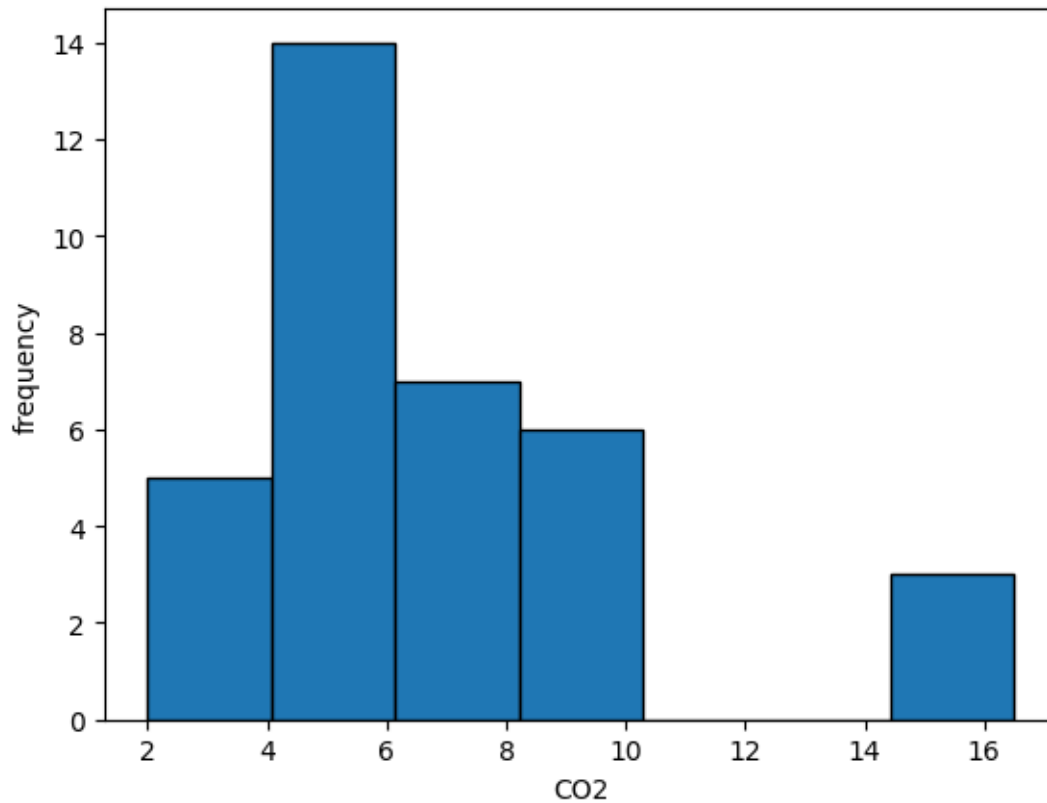
1.5 Problem # 1.10.

Analyze the `Carbon_West` (http://stat4ds.rwth-aachen.de/data/Carbon_West.dat) data file at the book's website by (a) constructing a frequency distribution and a histogram, (b) finding the mean, median, and standard deviation. Interpret each.

(a)

```
[ ]: carbon = pd.read_csv('https://stat4ds.rwth-aachen.de/data/Carbon_West.dat',
    ↪sep=r'\s+')
carbonFreqCount = carbon['CO2'].value_counts()
print(carbonFreqCount)
plt.hist(carbon['CO2'], bins='auto', edgecolor='black')
plt.xlabel("CO2")
plt.ylabel("frequency")
plt.show()
```

```
C02
6.2      3
5.3      3
4.3      3
3.5      2
5.9      2
6.9      1
15.4     1
15.1     1
4.0      1
2.0      1
8.3      1
8.7      1
9.2      1
8.9      1
4.6      1
7.3      1
4.4      1
5.4      1
3.6      1
9.9      1
7.7      1
9.3      1
5.7      1
5.0      1
4.5      1
6.5      1
16.5     1
Name: count, dtype: int64
```



(b) Your answer goes here

```
[ ]: carbon.describe()
```

```
[ ]:
      CO2
count  35.000000
mean    6.717143
std     3.356949
min     2.000000
25%     4.450000
50%     5.900000
75%     8.000000
max    16.500000
```

The mean of 6.7 is higher than the median 5.9 which means this distribution is skewed to the right because of the outliers (15.1, 15.4, and 16.5). The standard deviation 3.4 which is half the mean mostly due to the outliers.

1.6 Problem # 1.11.

According to Statistics Canada, for the Canadian population having income in 2019, annual income had a median of \$35,000 and mean of \$46,700. What would you predict about the shape of the

distribution? Why?

Your answer goes here

Skewed to the right since the mean is larger than the median which is likely due to an outlier or outliers

1.7 Problem # 1.13.

A report indicates that public school teacher's annual salaries in New York city have an approximate mean of \$69,000 and standard deviation of \$6,000. If the distribution has approximately a bell shape, report intervals that contain about (a) 68%, (b) 95%, (c) all or nearly all salaries. Would a salary of \$100,000 be unusual? Why?

Your answer goes here

- a) 68% would fall within 1 standard deviation
- b) 95% would fall within 2 standard deviation
- c) all or nearly all would fall within 3 standard deviations

A salary of \$100,000 would be unusual because it would be about 5 standard deviations from the mean which is outside the 3 standard deviations that contain all or nearly all salaries

1.8 Problem # 1.17.

From the Murder data file (<http://stat4ds.rwth-aachen.de/data/Murder.dat>) at the book's website, use the variable murder, which is the murder rate (per 100,000 population) for each state in the U.S. in 2017 according to the FBI Uniform Crime Reports. At first, do not use the observation for D.C. (DC). Using software:

- (a) Find the mean and standard deviation and interpret their values.
- (b) Find the five-number summary, and construct the corresponding box plot. Interpret.
- (c) Now include the observation for D.C. What is affected more by this outlier: The mean or the median? The range or the inter-quartile range?

Answer:

(a) Your answer goes here

```
[ ]: murder = pd.read_csv('http://stat4ds.rwth-aachen.de/data/Murder.dat',  
    ↪sep=r'\s+')  
murderNoDC = murder[murder['state'] != 'DC']  
murderNoDC.describe()
```

```
[ ]:      murder  
count    50.000000  
mean      4.874000  
std       2.586291  
min       1.000000  
25%       2.625000  
50%       4.850000  
75%       6.175000
```

max 12.400000

The average murder rate per 100,000 population is about 4.87 and the observations deviate by 2.59 from the average. The max is an outlier at 12.4 which is about 3 standard deviations above the mean.

(b) Your answer goes here

We found the 5 number summary in part a so we'll just construct the boxplot using that data:

min: 1

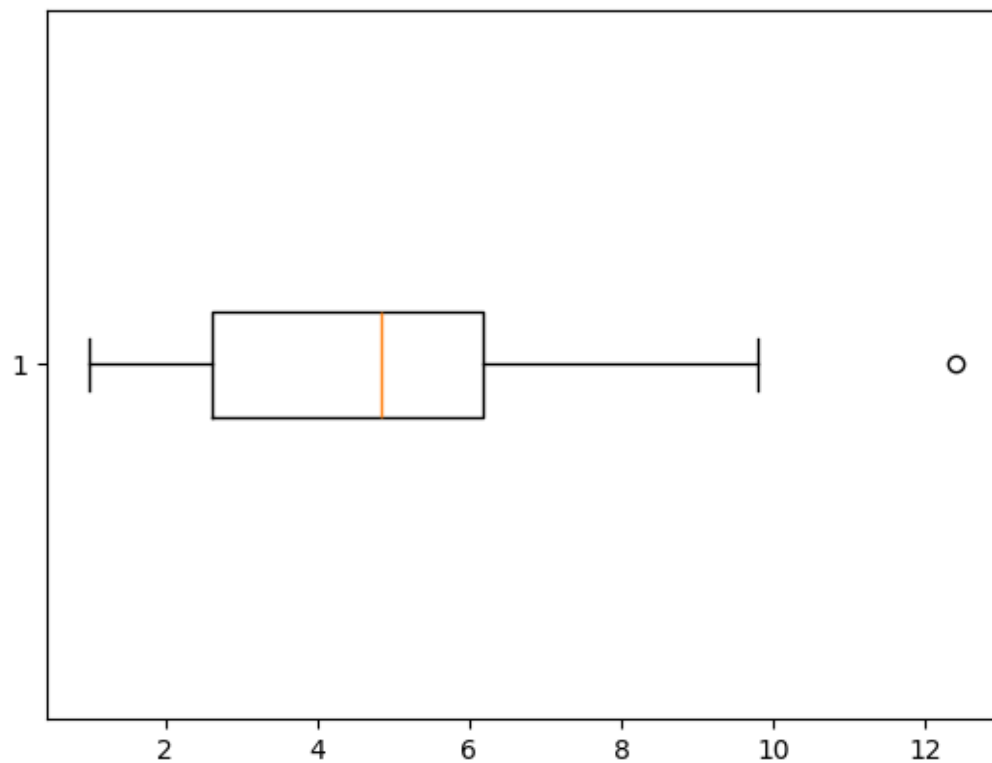
Q1: 2.63

median: 4.85

Q3: 6.18

max: 12.4

```
[ ]: plt.boxplot(murderNoDC['murder'], vert=False)
plt.show()
```



We can see that the data is skewed right since an outlier point (12.5) lie beyond the right whisker and there is no similar outliers on the left side.

(c) Your answer goes here


```
[ ]: murder.describe()
```

```
[ ]:      murder
count    51.000000
mean      5.252941
std       3.725391
min       1.000000
25%       2.650000
50%       5.000000
75%       6.450000
max      24.200000
```

The mean and the range are more affected by the outlier as we can see a greater difference in the observations when including DC in those categories than in median and IQR

1.9 Problem # 1.18.

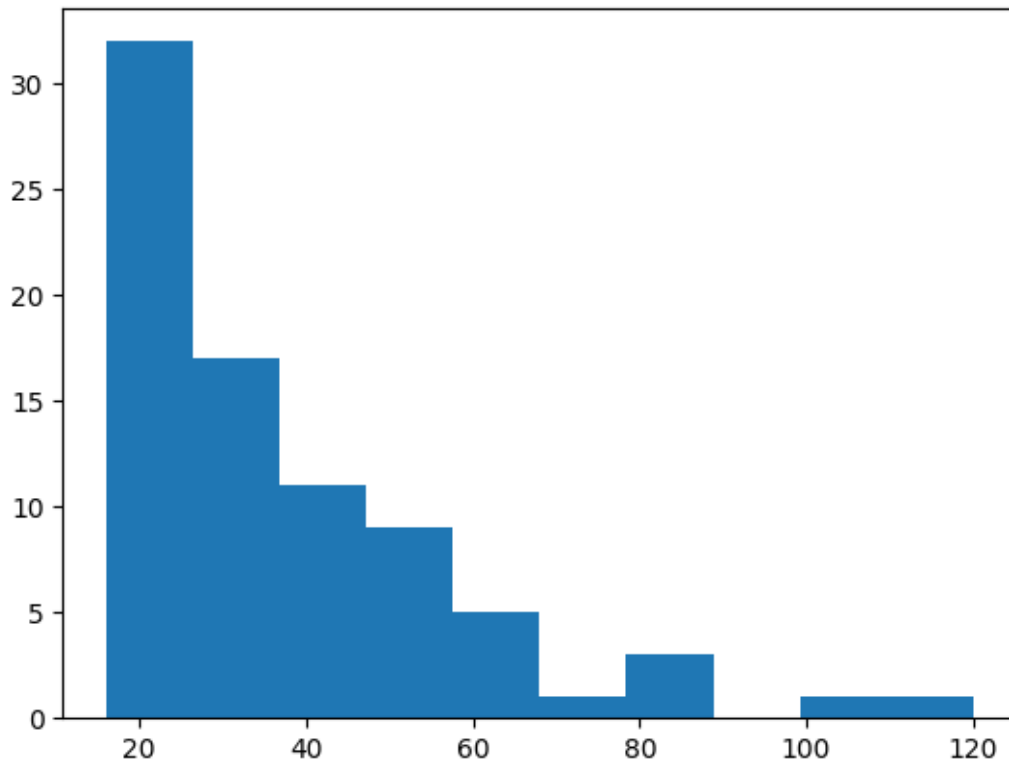
The Income data file (<http://stat4ds.rwth-aachen.de/data/Income.dat>) at the book's website reports annual income values in the U.S., in thousands of dollars.

- (a) Using software, construct a histogram. Describe its shape.
- (b) Find descriptive statistics to summarize the data. Interpret them.
- (c) The kernel density estimation method finds a smooth-curve approximation for a histogram. At each value, it takes into account how many observations are nearby and their distance, with more weight given those closer. Increasing the bandwidth increases the influence of observations further away. Plot a smooth-curve approximation for the histogram of income values. Summarize the impact of increasing and of decreasing the bandwidth substantially from the default value.
- (d) Construct and interpret side-by-side box plots of income by race (B = Black, H = Hispanic, W = White). Compare the incomes using numerical descriptive statistics

Answer:

(a) Your answer goes here

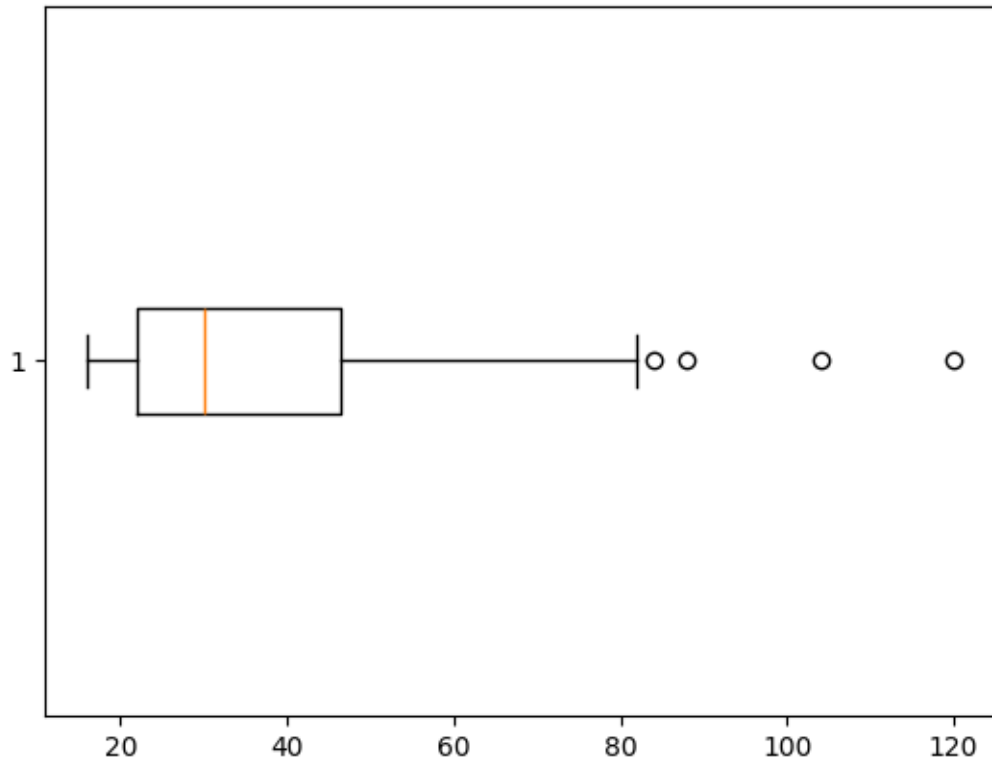
```
[ ]: income = pd.read_csv('http://stat4ds.rwth-aachen.de/data/Income.dat',
    ↪sep=r'\s+')
plt.hist(income['income'])
plt.show()
```



The distribution is skewed to the right

(b) Your answer goes here

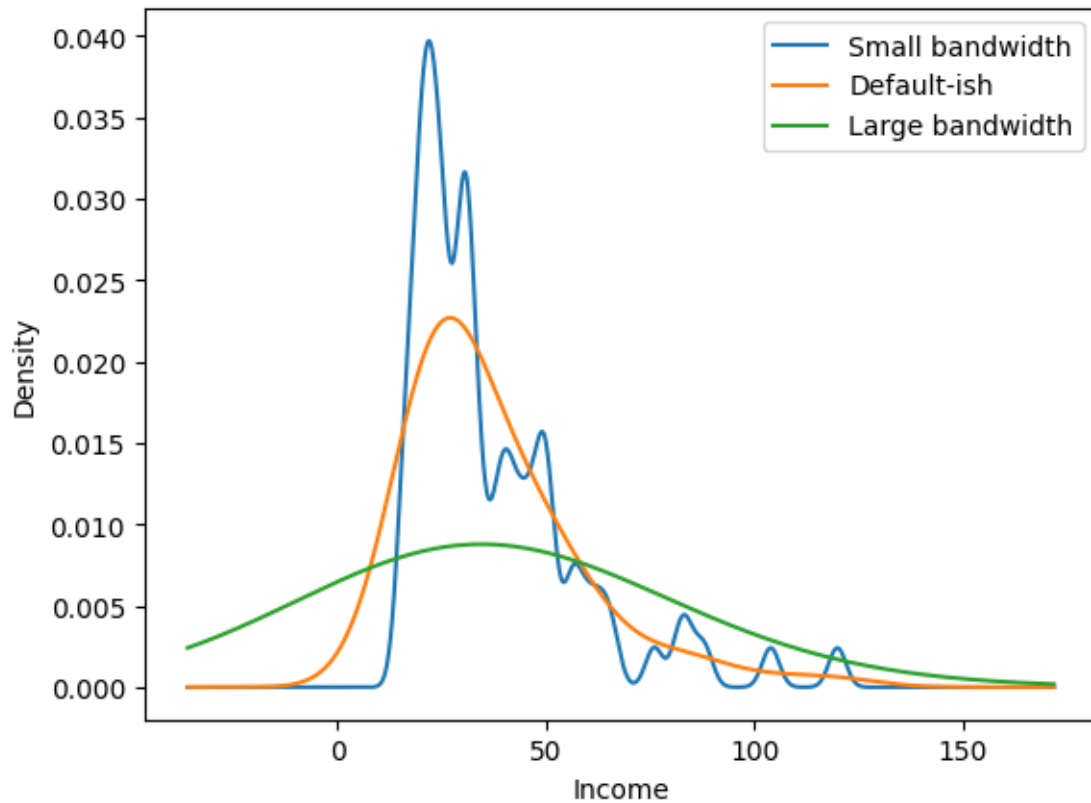
```
[ ]: plt.boxplot(income['income'], vert=False)
plt.show()
```



(c) Your answer goes here

Use Claude here to show me how to code kde

```
[ ]: income['income'].plot.kde(bw_method=0.1, label='Small bandwidth')
income['income'].plot.kde(bw_method=0.5, label='Default-ish')
income['income'].plot.kde(bw_method=2, label='Large bandwidth')
plt.legend()
plt.xlabel('Income')
plt.show()
```

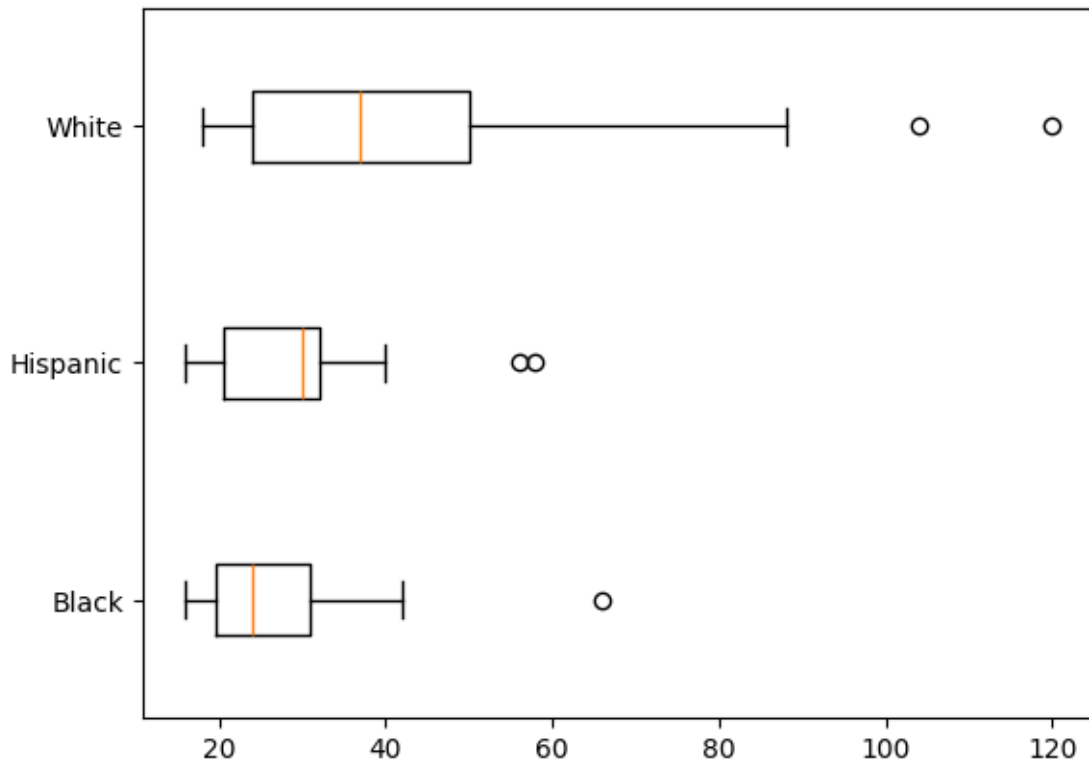


A smaller bandwidth makes each gaussian bump smaller so the total curve ends up looking jagged and a larger bandwidth makes each bump wider leading to a more stable curve.

(d) Your answer goes here

Used Claude here to help with side by side boxplots

```
[ ]: groups = [
    income[income['race'] == 'B']['income'],
    income[income['race'] == 'H']['income'],
    income[income['race'] == 'W']['income']
]
plt.boxplot(groups, tick_labels=['Black', 'Hispanic', 'White'], vert=False)
plt.show()
```



```
[ ]: income.groupby('race')['income'].describe()
```

```
[ ]:
```

	count	mean	std	min	25%	50%	75%	max
race								
B	16.0	27.75	13.284076	16.0	19.5	24.0	31.0	66.0
H	14.0	31.00	12.812254	16.0	20.5	30.0	32.0	58.0
W	50.0	42.48	22.869854	18.0	24.0	37.0	50.0	120.0

On average, white people have a higher income than black and hispanic people while also having a larger variance which we can see with the standard deviation in the income for white people being 22.87 and the median being 37 while the max is 120.

1.10 Problem # 1.19.

The Houses data file (<http://stat4ds.rwth-aachen.de/data/Houses.dat>) at the book's website lists the selling price (thousands of dollars), size (square feet), tax bill (dollars), number of bathrooms, number of bedrooms, and whether the house is new (1 = yes, 0 = no) for 100 home sales in Gainesville, Florida. Let's analyze the selling prices.

- Construct a frequency distribution and a histogram. Describe the shape.
- Find the percentage of observations that fall within one standard deviation of the mean. Why is this not close to 68%?

(c) Construct a box plot, and interpret.

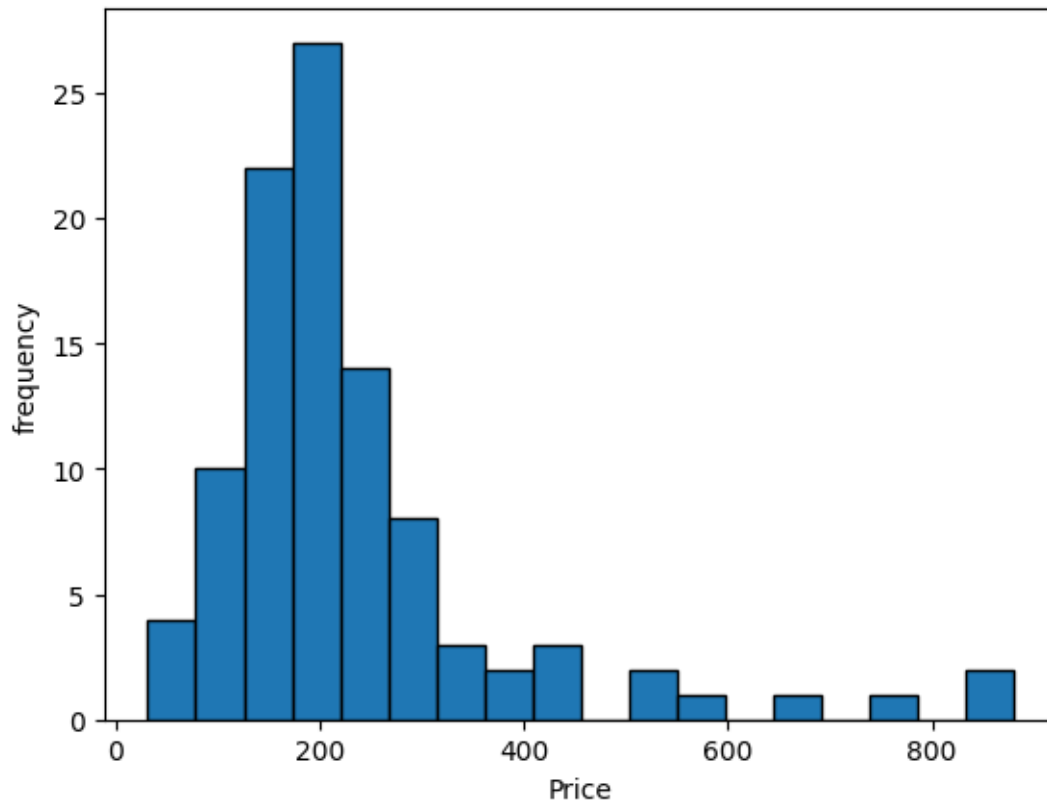
(d) Use descriptive statistics to compare selling prices according to whether the house is new.

Answer:

(a) Your answer goes here

```
[ ]: houses = pd.read_csv('http://stat4ds.rwth-aachen.de/data/Houses.dat',  
    ↪sep=r'\s+')  
print(houses['price'].value_counts())  
  
plt.hist(houses['price'], bins='auto', edgecolor='black')  
plt.xlabel("Price")  
plt.ylabel("frequency")  
plt.show()
```

```
price  
127.50    3  
150.00    3  
205.50    3  
210.00    3  
154.50    2  
..  
220.50    1  
294.75    1  
198.30    1  
132.60    1  
190.80    1  
Name: count, Length: 81, dtype: int64
```



This distributions is skewed to the right

(b) Your answer goes here

Asked Claude how to add a column to the dataset so I can set a boolean for if it falls within the first standard deviation. Then I'll do mean() on that column

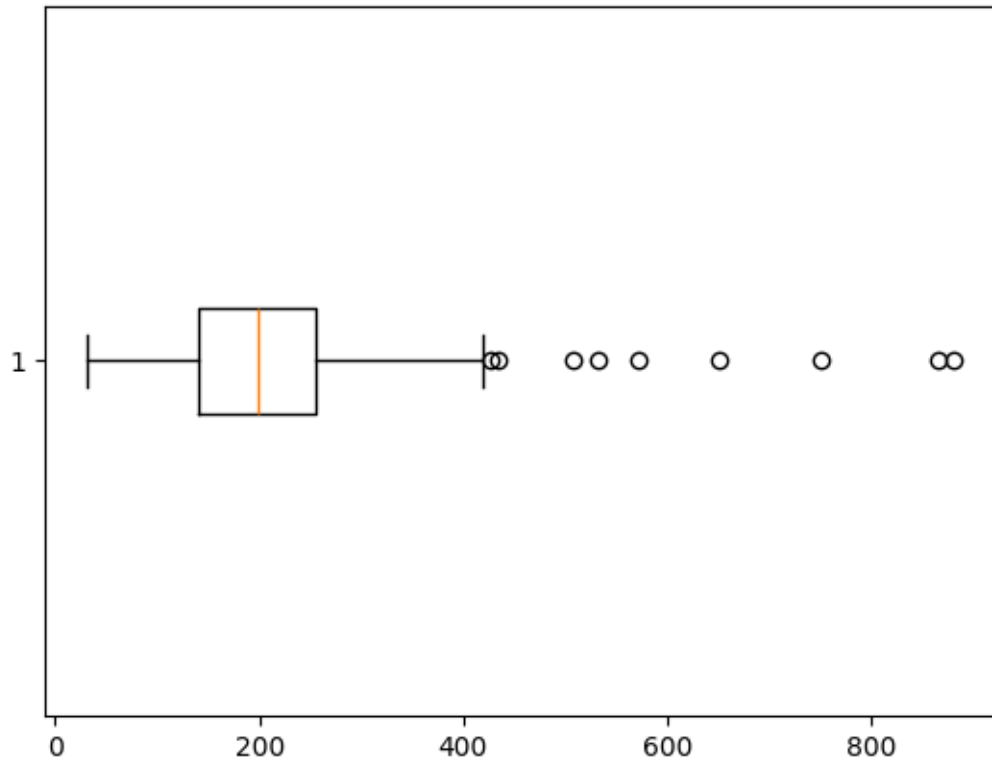
```
[ ]: center_of_gravity = houses['price'].mean()
      sd = houses['price'].std()
      houses['price_within_1sd'] = (houses['price'] >= center_of_gravity - sd) &
      ↪(houses['price'] <= center_of_gravity + sd)
      houses['price_within_1sd'].mean()
```

```
[ ]: np.float64(0.85)
```

85% of observations fall within 1 standard deviation

(c) Your answer goes here

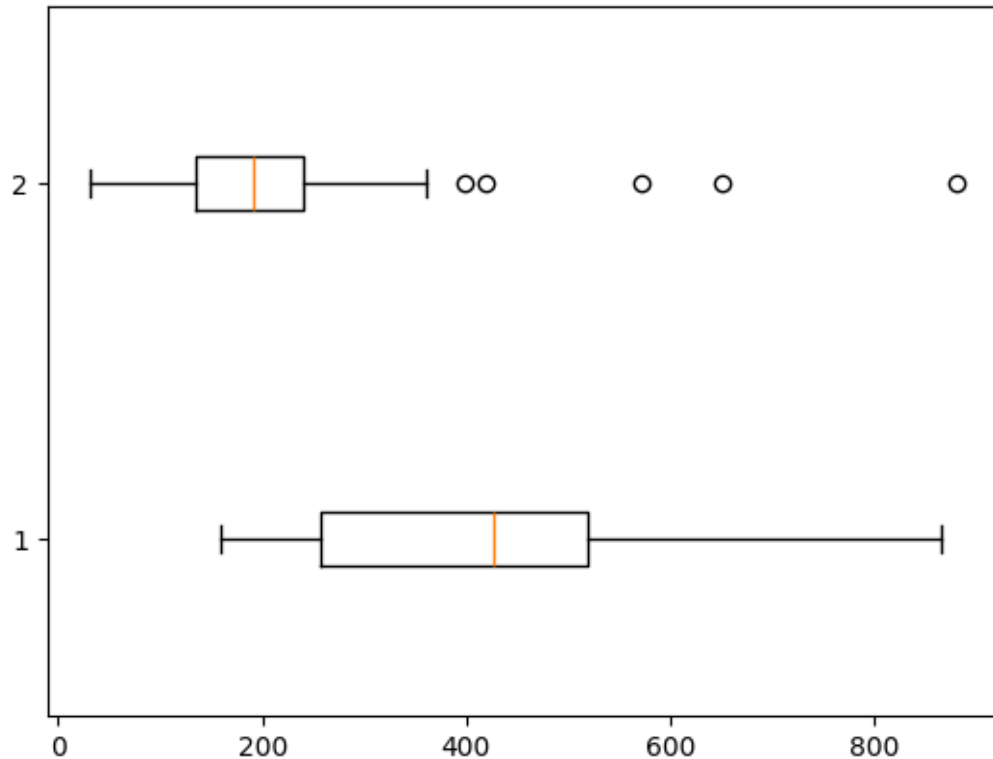
```
[ ]: plt.boxplot(houses['price'], vert=False)
      plt.show()
```



Distribution is skewed to the right with many outliers beyond the right whisker and you can see this represented by the median and IQR being in the lower price ranges while the tail is pulled to the right due to those outliers.

(d) Your answer goes here

```
[ ]: groups = [
    houses[houses['new'] == 1]['price'],
    houses[houses['new'] == 0]['price']
]
plt.boxplot(groups, vert=False)
plt.show()
```

The boxplot for the new houses show that it is not uncommon to see higher pricing for the houses when they are new. This is shown by the IQR sitting at a higher price than in the box plot for houses that aren't new. Then we have many outliers in the not new boxplot where as those outliers fall within the right whisker of the new houses boxplot

1.11 AI Problem: Preparing Data for Healthcare Predictions

AI in healthcare thrives on clean, insightful data prep: models like random forests predict gallstones from features like BMI and age, but garbage in = garbage out. Here, you'll perform EDA on the Gallstone-1 dataset to spot patterns, directly informing AI steps like feature scaling or class balancing. This is your first taste of how stats prevent biased predictions in tools saving lives, like FDA-approved AI diagnostics flagging risks pre-surgery.

1.11.1 Dataset

UCI's Gallstone-1 [<https://www.kaggle.com/datasets/xixama/gallstone-dataset-uci?resource=download>]. Load gallstone.xlsx (~320 rows).

1.11.2 Tasks

- Feature Exploration Load data. Show first 5 rows and info. Summarize key features and their distributions. Explain what these findings reveal about the data's central tendencies and variability, and how they might inform initial AI preprocessing decisions, such as normalization.

- (b) Exploratory Data Analysis Conduct a deeper analysis on key features and their relationship to `gallstone status`. Prepare a brief report on your findings. Include at least 5 clean, well-formatted visualizations.
- (c) Data Augmentation Generate 50 synthetic “augmented” rows for gallstone-present cases. Append, re-run your analysis in part (b). How does augmentation smooth variances, mimicking real GANs for underrepresented classes?

Answer:

(a) Your answer goes here

I’ve never done anything like this before and this wasn’t covered in the textbook so I had to ask Claude to walk me through how to do something like this.

```
[ ]: import kagglehub

path = kagglehub.dataset_download("xixama/gallstone-dataset-uci")

gallstones = pd.read_excel(f"{path}/gallstone.xlsx")
gallstones.head()
```

Using Colab cache for faster access to the 'gallstone-dataset-uci' dataset.

```
[ ]: Gallstone Status  Age  Gender  Comorbidity  Coronary Artery Disease (CAD) \
0          0      50      0          0          0
1          0      47      0          1          0
2          0      61      0          0          0
3          0      41      0          0          0
4          0      42      0          0          0

Hypothyroidism  Hyperlipidemia  Diabetes Mellitus (DM)  Height  Weight \
0          0          0          0      185      92.8
1          0          0          0      176      94.5
2          0          0          0      171      91.1
3          0          0          0      168      67.7
4          0          0          0      178      89.6

... High Density Lipoprotein (HDL)  Triglyceride \
0 ...          40.0          134.0
1 ...          43.0          103.0
2 ...          43.0          69.0
3 ...          59.0          53.0
4 ...          30.0          326.0

Aspartat Aminotransferaz (AST)  Alanin Aminotransferaz (ALT) \
0          20.0          22.0
1          14.0          13.0
2          18.0          14.0
3          20.0          12.0
```

4 27.0 54.0

	Alkaline Phosphatase (ALP)	Creatinine	Glomerular Filtration Rate (GFR) \
0	87.0	0.82	112.47
1	46.0	0.87	107.10
2	66.0	1.25	65.51
3	34.0	1.02	94.10
4	71.0	0.82	112.47

	C-Reactive Protein (CRP)	Hemoglobin (HGB)	Vitamin D
0	0.0	16.0	33.0
1	0.0	14.4	25.0
2	0.0	16.2	30.2
3	0.0	15.4	35.4
4	0.0	16.8	40.6

[5 rows x 39 columns]

```
[ ]: gallstones.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 319 entries, 0 to 318
```

```
Data columns (total 39 columns):
```

#	Column	Non-Null Count	Dtype
0	Gallstone Status	319 non-null	int64
1	Age	319 non-null	int64
2	Gender	319 non-null	int64
3	Comorbidity	319 non-null	int64
4	Coronary Artery Disease (CAD)	319 non-null	int64
5	Hypothyroidism	319 non-null	int64
6	Hyperlipidemia	319 non-null	int64
7	Diabetes Mellitus (DM)	319 non-null	int64
8	Height	319 non-null	int64
9	Weight	319 non-null	float64
10	Body Mass Index (BMI)	319 non-null	float64
11	Total Body Water (TBW)	319 non-null	float64
12	Extracellular Water (ECW)	319 non-null	float64
13	Intracellular Water (ICW)	319 non-null	float64
14	Extracellular Fluid/Total Body Water (ECF/TBW)	319 non-null	float64
15	Total Body Fat Ratio (TBFR) (%)	319 non-null	float64
16	Lean Mass (LM) (%)	319 non-null	float64
17	Body Protein Content (Protein) (%)	319 non-null	float64
18	Visceral Fat Rating (VFR)	319 non-null	int64
19	Bone Mass (BM)	319 non-null	float64
20	Muscle Mass (MM)	319 non-null	float64
21	Obesity (%)	319 non-null	float64
22	Total Fat Content (TFC)	319 non-null	float64

23	Visceral Fat Area (VFA)	319	non-null	float64
24	Visceral Muscle Area (VMA) (Kg)	319	non-null	float64
25	Hepatic Fat Accumulation (HFA)	319	non-null	int64
26	Glucose	319	non-null	float64
27	Total Cholesterol (TC)	319	non-null	float64
28	Low Density Lipoprotein (LDL)	319	non-null	float64
29	High Density Lipoprotein (HDL)	319	non-null	float64
30	Triglyceride	319	non-null	float64
31	Aspartat Aminotransferaz (AST)	319	non-null	float64
32	Alanin Aminotransferaz (ALT)	319	non-null	float64
33	Alkaline Phosphatase (ALP)	319	non-null	float64
34	Creatinine	319	non-null	float64
35	Glomerular Filtration Rate (GFR)	319	non-null	float64
36	C-Reactive Protein (CRP)	319	non-null	float64
37	Hemoglobin (HGB)	319	non-null	float64
38	Vitamin D	319	non-null	float64

dtypes: float64(28), int64(11)

memory usage: 97.3 KB

Looked up which factors lead to higher risk of gallstones

Key feature columns when predicting gallstones:

BMI, Age, Gender, Total Cholesterol, Obesity

```
[ ]: key_features = [
    'Body Mass Index (BMI)',
    'Age',
    'Gender',
    'Total Cholesterol (TC)',
    'Obesity (%)'
]
```

Gallstone positive

```
[ ]: gallstones[gallstones['Gallstone Status'] == 1][key_features].describe()
```

	Body Mass Index (BMI)	Age	Gender	Total Cholesterol (TC)	\
count	158.000000	158.000000	158.000000	158.000000	
mean	29.527848	48.512658	0.569620	204.145570	
std	5.274610	11.200397	0.496704	48.278462	
min	18.800000	20.000000	0.000000	88.000000	
25%	26.025000	39.250000	0.000000	173.000000	
50%	29.150000	50.000000	1.000000	198.000000	
75%	32.475000	56.000000	1.000000	230.750000	
max	48.200000	76.000000	1.000000	360.000000	

	Obesity (%)
count	158.000000

```

mean      41.817089
std       154.507974
min        1.500000
25%       13.900000
50%       25.165000
75%       37.710000
max       1954.000000

```

Gallstone negative

```
[ ]: gallstones[gallstones['Gallstone Status'] == 0][key_features].describe()
```

```
[ ]:
      Body Mass Index (BMI)      Age      Gender  Total Cholesterol (TC) \
count      161.000000  161.000000  161.000000      161.000000
mean        28.238509   47.633540    0.416149    202.857143
std          5.290523   12.969527    0.494457     43.284792
min         17.400000   20.000000    0.000000     60.000000
25%         24.800000   38.000000    0.000000    172.000000
50%         27.700000   49.000000    0.000000    199.000000
75%         31.400000   56.000000    1.000000    234.000000
max         49.700000   96.000000    1.000000    325.000000

```

```

      Obesity (%)
count      161.000000
mean        29.994348
std         21.598417
min          0.400000
25%         14.100000
50%         25.800000
75%         43.000000
max        125.600000

```

I'll compare the same variables on the two tables starting from BMI and going right. The mean BMI for positive gallstone status is 29.5 which is slightly higher than the mean BMI for the gallstone negative group (28.24) but BMI does not take into account the amount of muscle mass a subject has so we look at Obesity, skipping the other features for now, and see that the mean obesity for the group that is gallstone positive is 41.82 which is higher than the mean obesity of the gallstone negative group (29.99), however, this gap in mean obesity is due to the outlier in the positive group (1954) which is odd considering that the Obesity column is measured in percentage.

The mean age between the two groups are similar with about a .9 year gap (48.51 for the positive group and 47.63 for the negative group) so age doesn't seem to factor into whether they have gallstones although we do need to note that there is an outlier in the negative group which moves the mean age up by a little bit but if we don't include that then the difference in mean age between the two groups still isn't a large enough gap to make a reasonable connection between age and gallstone status.

In the positive group there is a larger percentage of females that are gallstone positive (57%) and in the negative group there is a larger percentage of males that are gallstone negative (58%).

The mean total cholesterol for the positive group is 204.15 close to the number that was observed in the negative group (202.86). The same can be said about the other measurements from describe() for this column.

There was no notable difference in standard deviation in any column except for Obesity largely due to the 1954 outlier but if we look at the Q1, median, and Q3 values and compare them with the same values from the negative group we can see that the obesity percentages are actually lower in the for those values in the positive group.

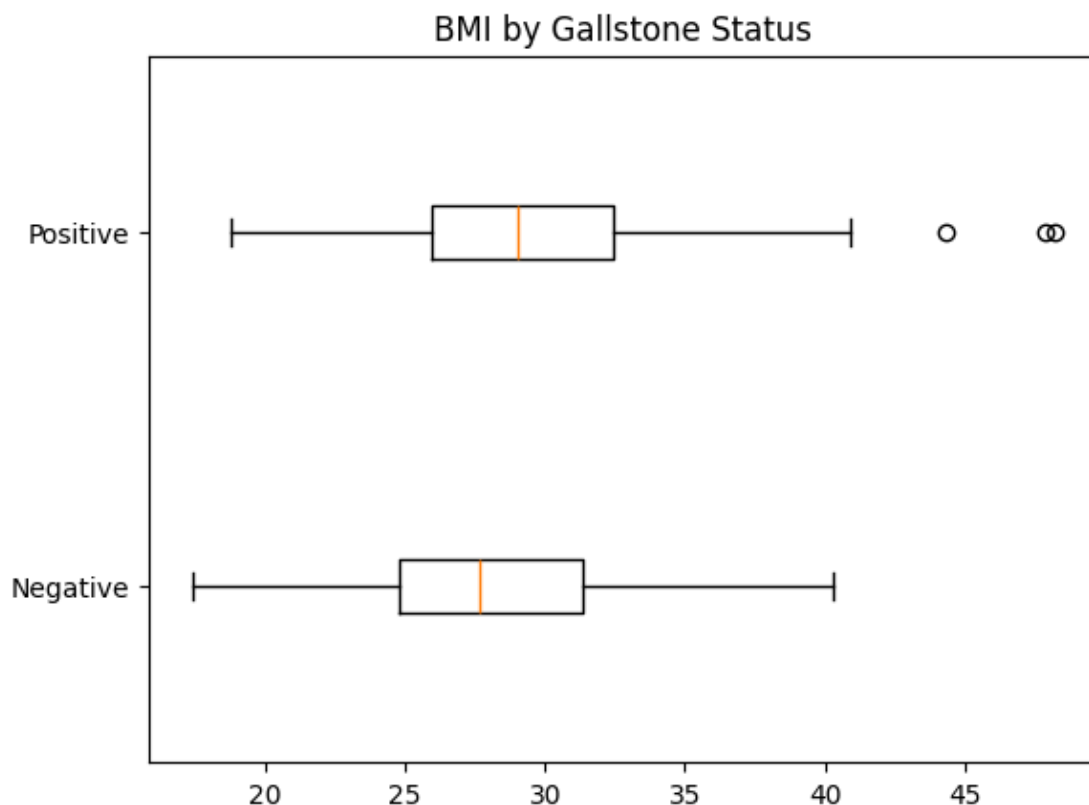
The categories have different scales so something like cholesterol would be weighed more heavily compared to BMI but if normalization were applied to this data it would distort the statistics due to the obesity outlier so this data needs to be cleaned first.

(b) Your answer goes here

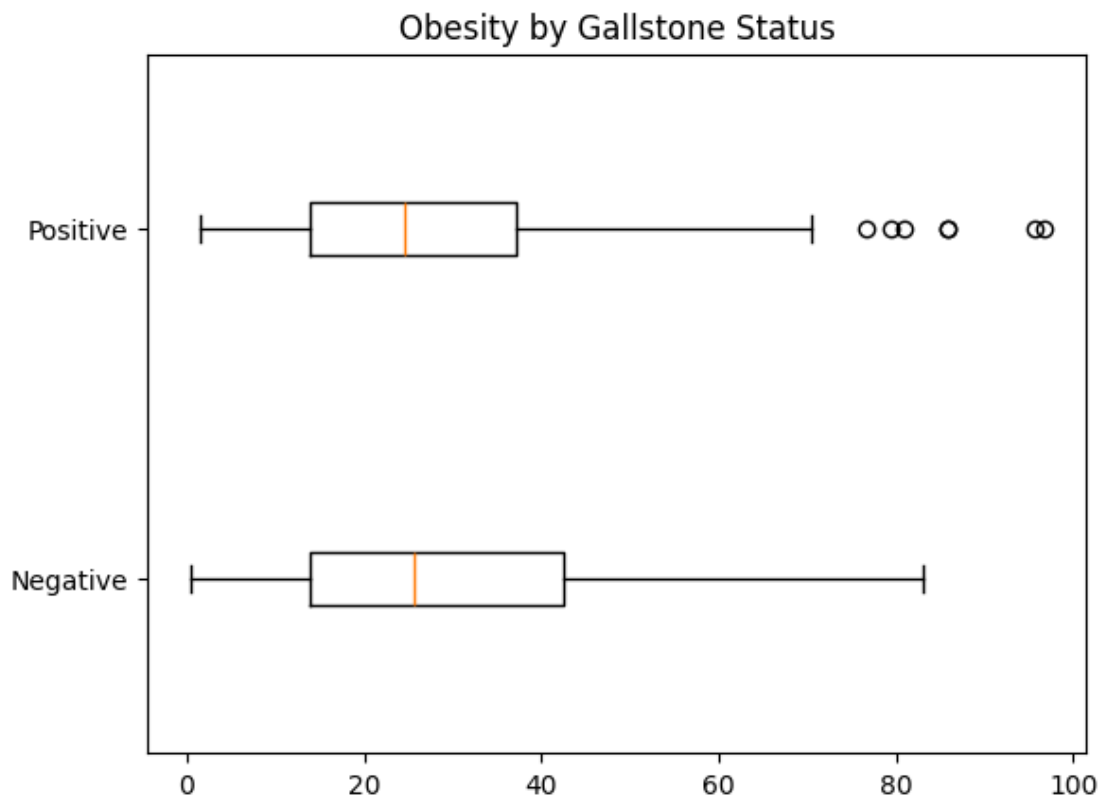
Choosing to drop the row with the obesity outlier so that the data is cleaner

```
[ ]: gallstones = gallstones[gallstones['Obesity (%)'] <= 100].reset_index(drop=True)
negative = gallstones['Gallstone Status'] == 0
positive = gallstones['Gallstone Status'] == 1

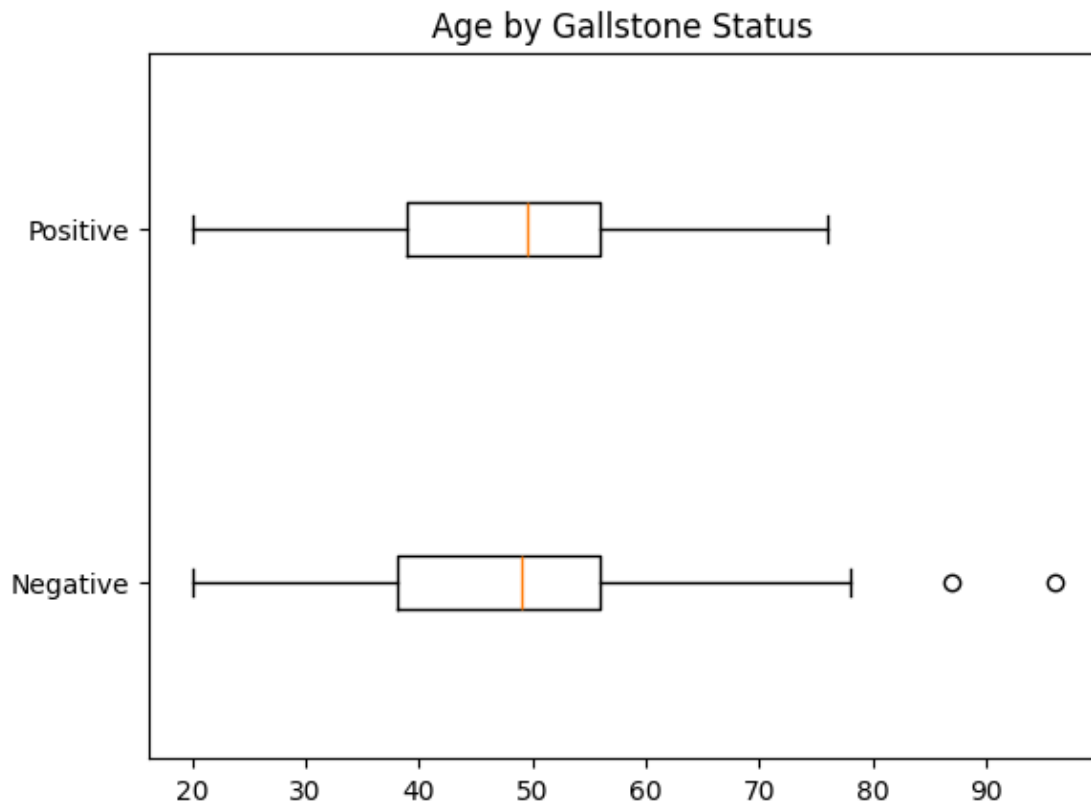
plt.boxplot([gallstones[negative]['Body Mass Index (BMI)'],
             ↪gallstones[positive]['Body Mass Index (BMI)']], vert=False,
             ↪tick_labels=['Negative', 'Positive'])
plt.title('BMI by Gallstone Status')
plt.show()
```



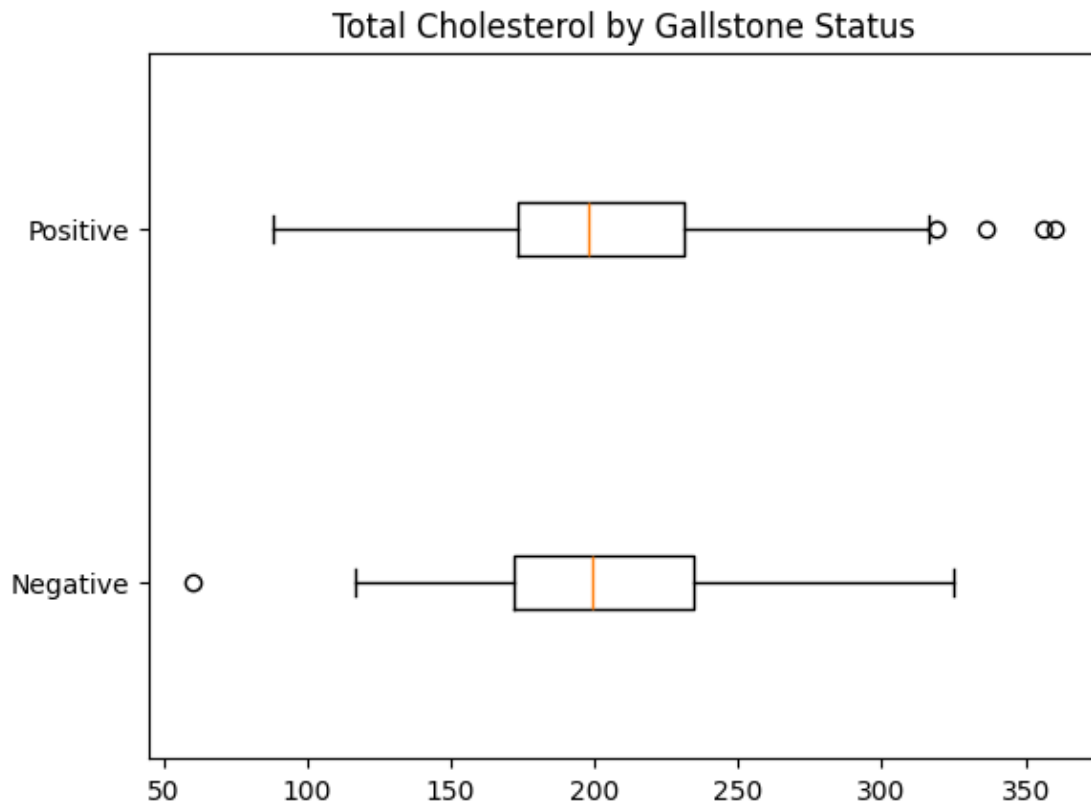
```
[ ]: plt.boxplot([gallstones[negative]['Obesity (%)'], gallstones[positive]['Obesity_
↪(%)']], vert=False, tick_labels=['Negative', 'Positive'])
plt.title('Obesity by Gallstone Status')
plt.show()
```



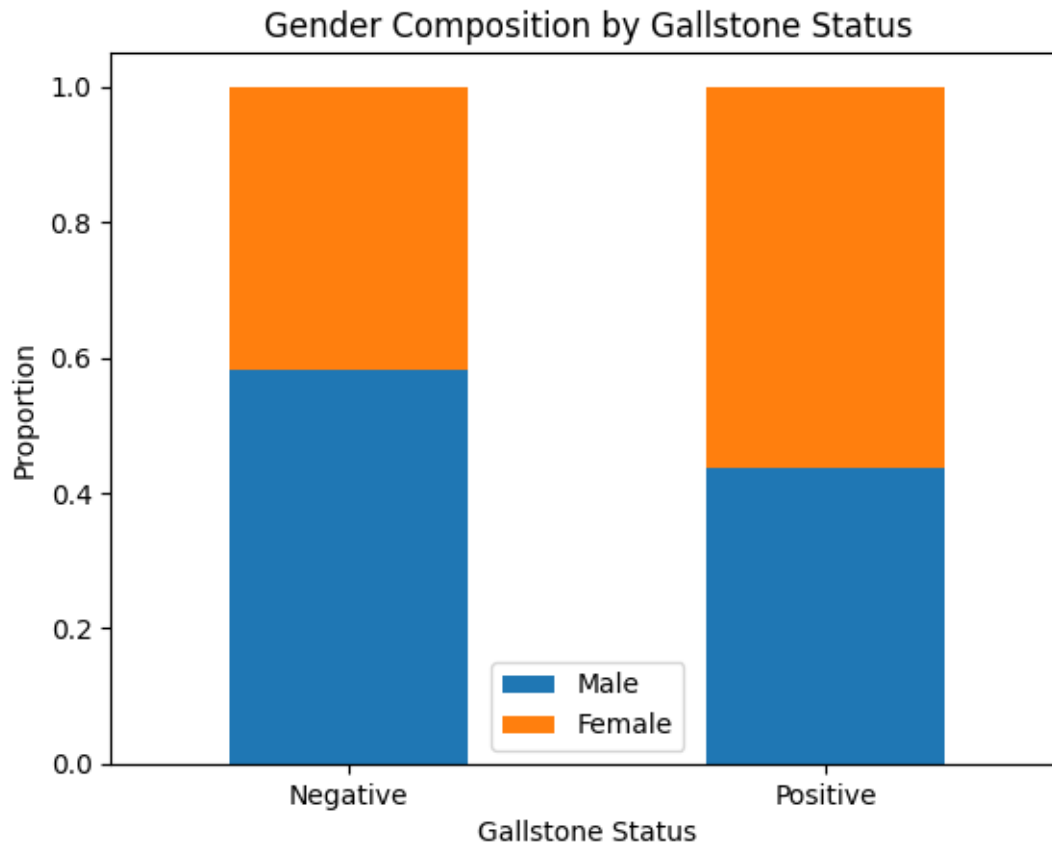
```
[ ]: plt.boxplot([gallstones[negative]['Age'], gallstones[positive]['Age']],
↪vert=False, tick_labels=['Negative', 'Positive'])
plt.title('Age by Gallstone Status')
plt.show()
```



```
[ ]: plt.boxplot([gallstones[negative]['Total Cholesterol (TC)'],  
                ↪gallstones[positive]['Total Cholesterol (TC)'], vert=False,  
                ↪tick_labels=['Negative', 'Positive'])  
plt.title('Total Cholesterol by Gallstone Status')  
plt.show()
```

```
[ ]: pd.crosstab(gallstones['Gallstone Status'], gallstones['Gender'],
    ↪normalize='index').plot(kind='bar', stacked=True)
plt.title('Gender Composition by Gallstone Status')
plt.xticks(ticks=[0, 1], labels=['Negative', 'Positive'], rotation=0)
plt.xlabel('Gallstone Status')
plt.ylabel('Proportion')
plt.legend(['Male', 'Female'])
plt.show()
```



The distributions for each feature have notable overlap in the box plot ranges and doesn't support these features as discriminating features for gallstone status, however, the BMI for the positive distributions does show a rightward shift. Overall, the features that show the most separation between the two groups are gender, BMI, and obesity while age shows little to no relationship.

(c) Your answer goes here

Asked claude to help me with generating the synthetic rows

```
[ ]: print(gallstones['Gallstone Status'].value_counts())
numeric_cols = gallstones.select_dtypes(include='number').columns.
    drop(['Gallstone Status', 'Gender'])
synthetic = gallstones[positive].sample(50, replace=True, random_state=42).
    copy()
```

Gallstone Status

0 160

1 156

Name: count, dtype: int64

Add noise here

```
[ ]: for col in numeric_cols:
    noise = np.random.normal(0, gallstones[positive][col].std() * 0.05, size=50)
    synthetic[col] = synthetic[col].values + noise
augmented_gallstones = pd.concat([gallstones, synthetic], ignore_index=True)
print(augmented_gallstones['Gallstone Status'].value_counts())
```

Gallstone Status

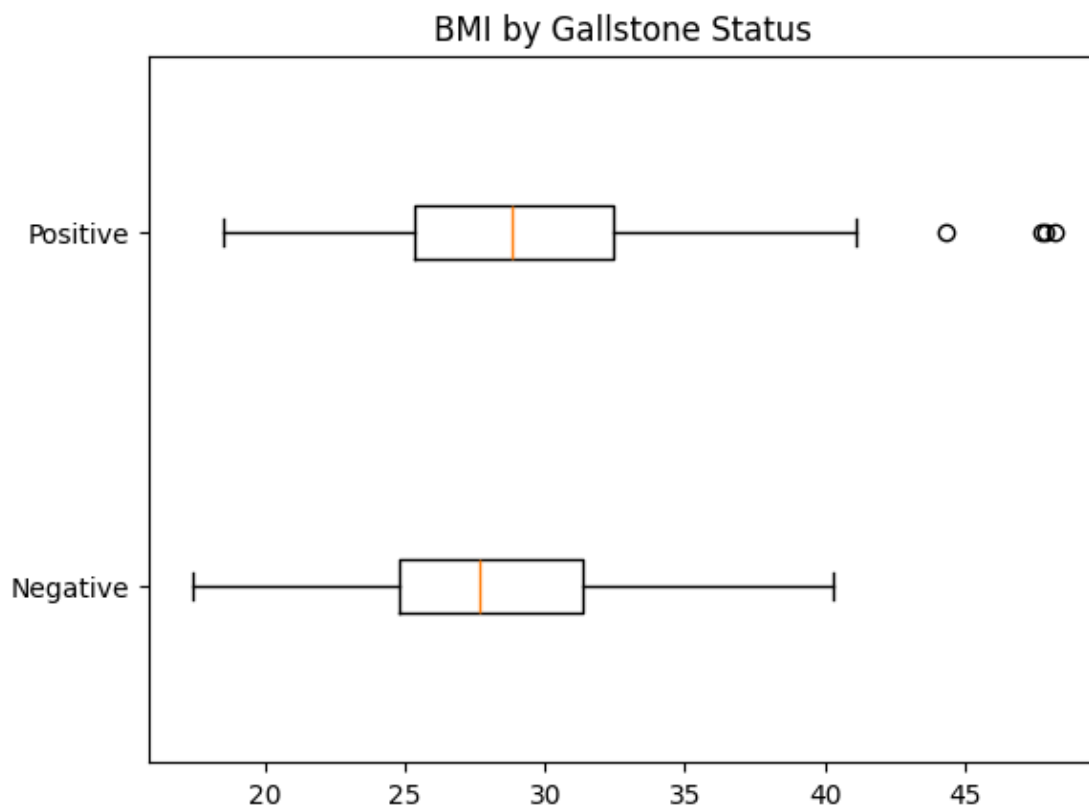
1 206

0 160

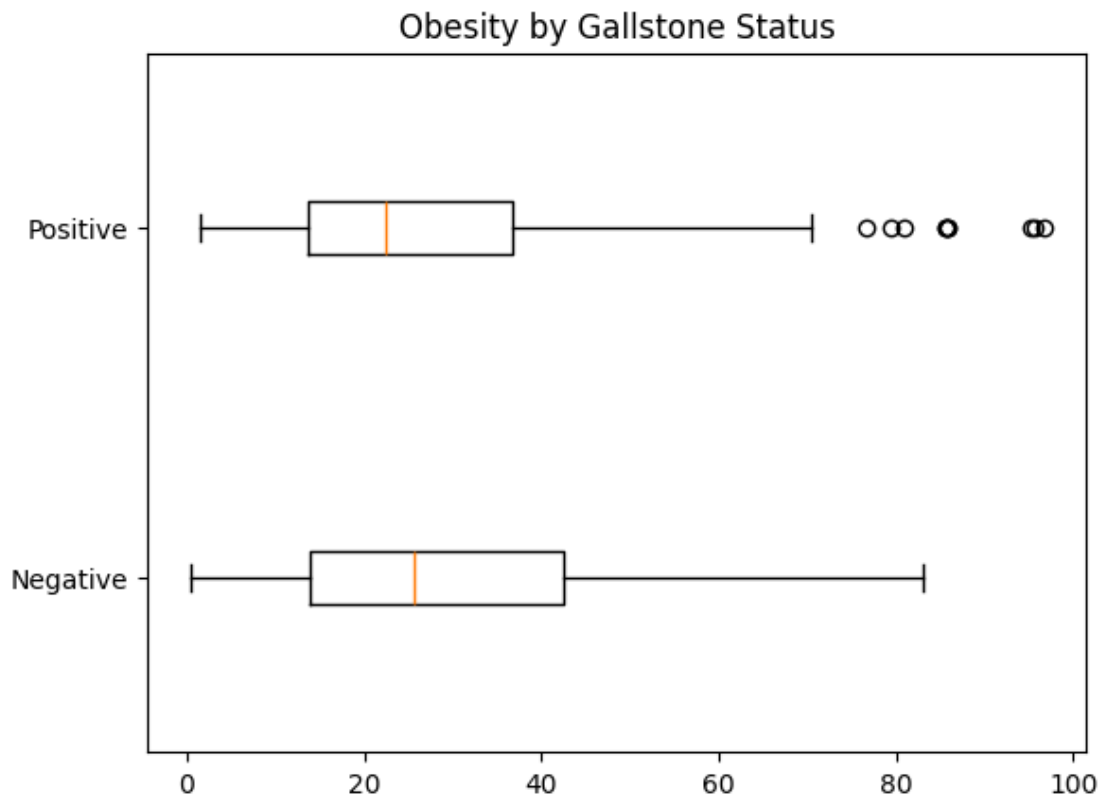
Name: count, dtype: int64

```
[ ]: augmented_positive = augmented_gallstones['Gallstone Status'] == 1
augmented_negative = augmented_gallstones['Gallstone Status'] == 0
```

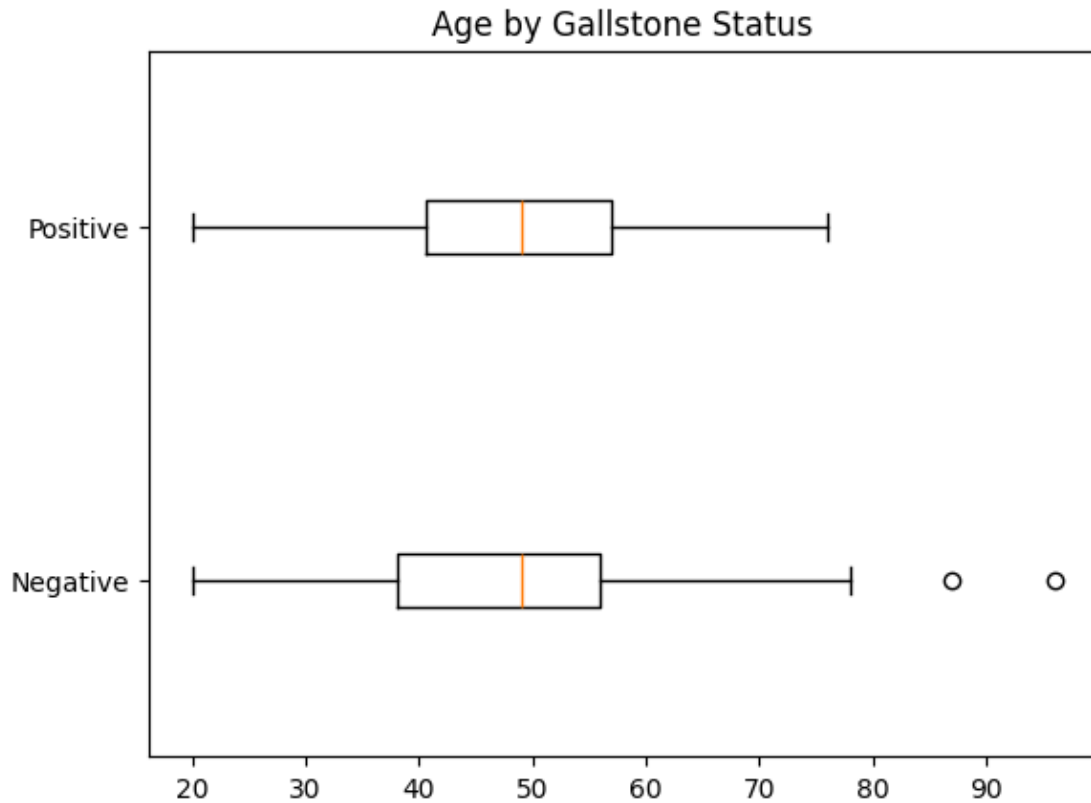
```
[ ]: plt.boxplot([augmented_gallstones[augmented_negative]['Body Mass Index (BMI)'],
    augmented_gallstones[augmented_positive]['Body Mass Index (BMI)'],
    vert=False, tick_labels=['Negative', 'Positive'])
plt.title('BMI by Gallstone Status')
plt.show()
```



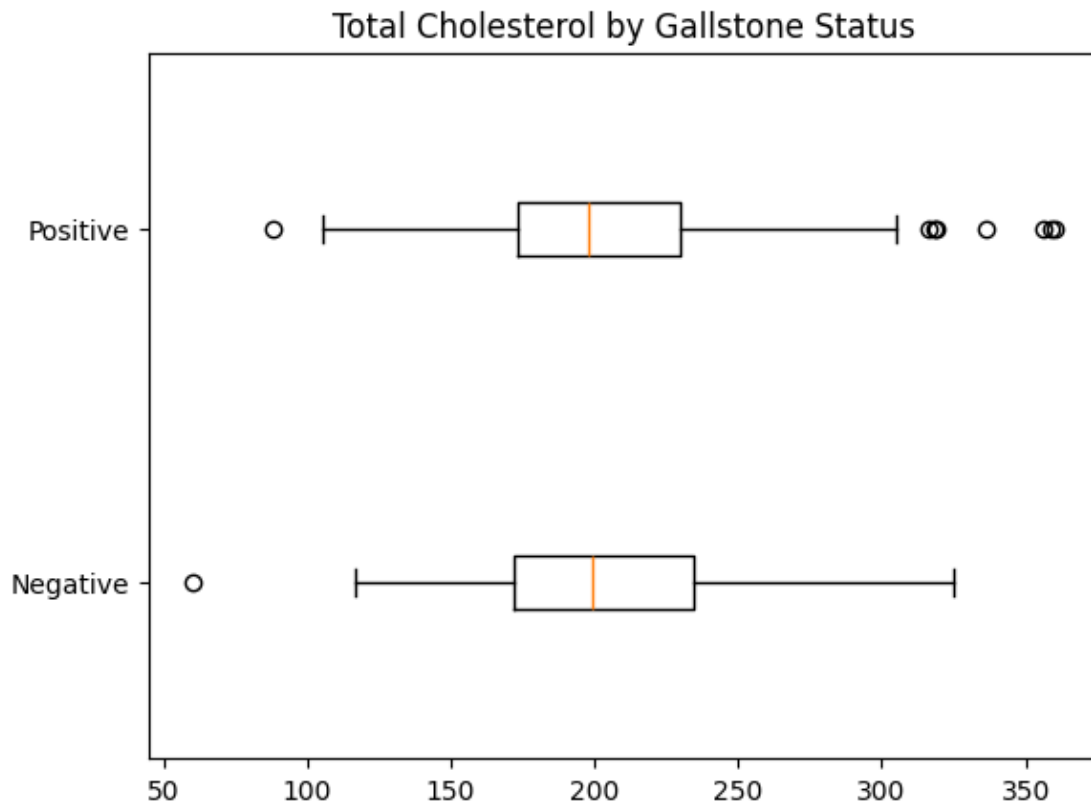
```
[ ]: plt.boxplot([augmented_gallstones[augmented_negative]['Obesity (%)'],  
    ↳ augmented_gallstones[augmented_positive]['Obesity (%)']], vert=False,  
    ↳ tick_labels=['Negative', 'Positive'])  
plt.title('Obesity by Gallstone Status')  
plt.show()
```



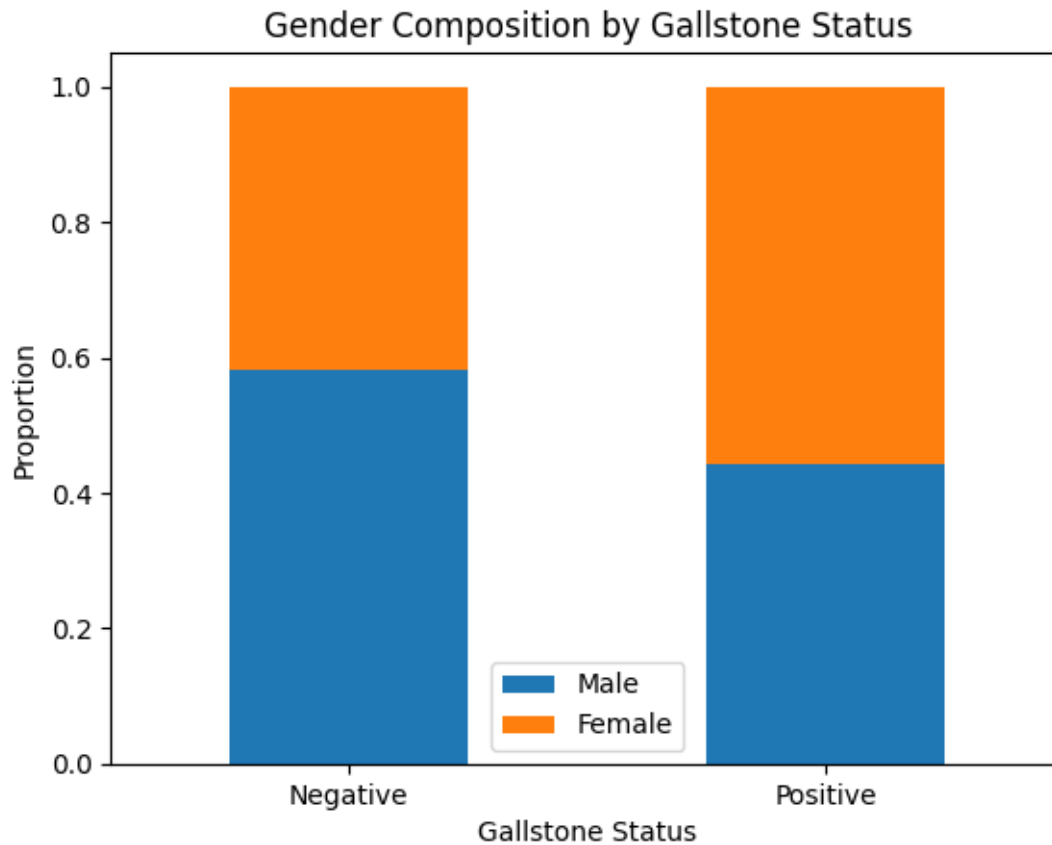
```
[ ]: plt.boxplot([augmented_gallstones[augmented_negative]['Age'],  
    ↳ augmented_gallstones[augmented_positive]['Age']], vert=False,  
    ↳ tick_labels=['Negative', 'Positive'])  
plt.title('Age by Gallstone Status')  
plt.show()
```



```
[ ]: plt.boxplot([augmented_gallstones[augmented_negative]['Total Cholesterol_
    ↪(TC)'], augmented_gallstones[augmented_positive]['Total Cholesterol (TC)']],
    ↪vert=False, tick_labels=['Negative', 'Positive'])
plt.title('Total Cholesterol by Gallstone Status')
plt.show()
```



```
[ ]: pd.crosstab(augmented_gallstones['Gallstone Status'],
    → augmented_gallstones['Gender'], normalize='index').plot(kind='bar',
    → stacked=True)
plt.title('Gender Composition by Gallstone Status')
plt.xticks(ticks=[0, 1], labels=['Negative', 'Positive'], rotation=0)
plt.xlabel('Gallstone Status')
plt.ylabel('Proportion')
plt.legend(['Male', 'Female'])
plt.show()
```



(b) answer here

Augmentation adds small noise around in the data within the feature's standard deviation similar to bandwidth in kde except adjusting bandwidth in kde doesnt change the data but data augmentation does.